MIL-W-50549A (MU)
12 June 1973
SUPERSEDING
MIL-W-50549 (MU)
10 July 1970

MILITARY SPECIFICATION

WARHEAD, 2.75 INCH ROCKET, FLECHETTE, WDU-4A/A, WDU-13/A, PARTS FOR

1. SCOPE

1.1 This specification covers the quality assurance provisions and special requirements not covered by the drawings for Warhead Flechette, 2.75 Inch Rocket, WDU-4A/A, WDU-13/A, Parts For.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for

STANDARDS

FEDERAL

FED-STD-151 - Metals, Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1168- Lot Numbering of Ammunition
MIL-STD-1169- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1340

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DRAWINGS

AIR FORCE

67D9700 - Warhead, 2.75 Inch Rocket, Flechette, WDU-4A/A

PICATINNY ARSENAL

9259995 - Warhead, 2.75 Inch Rocket, Flechette, 60 Grain

(Copies of specifications, standards, drawings and publications listed above or required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Warhead Assemblies.-The warhead assemblies shall comply with all requirements specified on Drawing (dwg.) 67D9700 (For 20 Grain Flechettes) and dwg. 9259995 (For 60 Grain Flechettes) and with all requirements specified in applicable specifications.

3.3 Metal Defects.-All of the components, prior to painting or chemical treatment, shall be free of surface imperfections on the interior and exterior surfaces of the part which are indications of discontinuities internal to the metal. (Included are cracks, lube bursts, pipes, laminations, segregations, porosity, flakes, inclusions and other internal defects).

3.4 Workmanship.-The components and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, unblended radii, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent. Surface coatings shall be continuous except for a few slight scratches not exposing base material. All required markings shall be neat and sharply defined.

3.5 First Article Inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 to define terms used herein. The provisions of Specification MIL- A-2550 shall apply.

4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization. A contractor shall not knowingly include defective units in a lot.

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4.1.2 Inspections.-Unless otherwise specified in the contract, the following types of inspection(s) shall be conducted:

a. First Article Inspection

b. Quality Conformance (Production Lot Acceptance) Inspection

4.2 First Article Inspection

4.2.1 Submission.-All components and items submitted in the first article sample shall have been produced by the contractor or furnished by a sub-contractor and shall have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract, and shall submit a record of this inspection with the sample, including statements of findings of tests and examinations, and certificates of conformance for materials (See 4.1.1c). A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply such as to significantly affect product uniformity as determined by the Government. The first article sample shall consist of five (5) sets of the component parts (unpainted) and five (5) unpainted assemblies. See paragraph 6.4 of this specification for instructions for submission.

4.2.2 Inspections to be Performed.-First article assemblies, components and test specimens will be subjected by the Government to any or all of the examinations and tests specified in this detail specification and to any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure in the sample to comply with any of the stated requirements.

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4.3 Quality Conformance Inspection (Production Lot Acceptance)

4.3.1 Lot Formation.-The term "inspection lot", as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in Standard MIL-STD-105. Changes to the process, specifications or drawings, not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall bear a unique lot number.

4.3.2 Examination.-The examinations listed in this detail specification for Classification of Defects (and tests) shall be performed on inspection lots as defined in 4.3.1 of this specification.

a. Sampling Plans.-Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105, except that variables sampling plans in accordance with MIL-STD-414 or continuous sampling plans in accordance with MIL-STD-1235 may be used if requested by the contractor and approved by the Government.

b. Critical Defects.-Unless otherwise specified, inspection for Critical Defects shall be 100 percent (%).

c. Major and Minor Defects.-When required, Major Defects shall be inspected 100%. Except where 100% inspection is specified, inspection for Major and Minor Defects shall be to the individual AQL indicated in the Classification of Defects and Test paragraphs provided in this detail specification. The sampling plans for testing (See paragraph 4.4) shall be as specified.

d. Unlisted Defects.-See MIL-A-2550

e. Disposition.-Unless otherwise specified, the disposition of defective units and rejected lots shall be in accordance with the requirements of MIL-STD-105.

f. Instead of applying the AQL's designated herein to the individual defects, the group AQL's listed for each Major or Minor subparagraph in 4.3.2 may be utilized at the option of the contractor for those groups where a group AQL is assigned. Each group AQL applies to a Major or Minor group indicated in the subparagraphs of paragraph 4.3.2, and if this method is elected, it shall be utilized for all of the subparagraphs of paragraph 4.3.2.

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.3.2.1 CASE, WARHEAD

DRAWING NUMBER 65D11036

COVERING DETAIL OF DWG. 67D9700

CATEGORY	CHARACTERISTICS/DEFECTS AND TESTS	DEFECT CODE NO.	AQL OR 100%	PARAGRAPH REFERENCE REQMT.	PARAGRAPH REFERENCE TEST	METHOD OF INSPECTION
CRITICAL: None defined						
MAJOR:		GROUP AQL 2.5%				
	Length of forward chamber	01001	0.40	Para. 3.2	N/A	Step Plug Gage
	Individual inside diameter forward chamber, maximum (max.)	01002	0.40	"	"	Dial Bore Gage
	Individual inside diameter forward chamber, minimum (min.)	01003	0.40	"	"	Dial Bore Gage
	Wall thickness, forward chamber	01004	0.40	"	"	Indicator Thickness Gage w/Mandrel
	Straightness outside diameter forward chamber	01005	0.40	"	"	Fixture w/Dial Indicator
	Length aft chamber	01006	0.40	"	"	Flush Pin Gage

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.3.2.1 CASE, WARHEAD

DRAWING NUMBER 65D11036

COVERING DETAIL OF DWG. 67D9700

CATEGORY	CHARACTERISTICS/DEFECTS AND TESTS	DEFECT CODE NO.	AQL OR 100%	PARAGRAPH REFERENCE REQMT.	PARAGRAPH REFERENCE TEST	METHOD OF INSPECTION
MAJOR:	Concentricity of aft section of case to outside diameter	01014	0.40	Para. 3.2	N/A	Concentricity Gage w/Dial Indicator
	Concentricity of aft section of case to pitch diameter of threads	01015	0.40	"	"	Concentricity Gage
	Major diameter of threads, min.	01016	0.40	"	"	Adjustable Snap Gage
	Pitch diameter of threads	01017	0.40	"	"	Thread Ring Gage GO/NOT GO
	Length from aft end to forward portion of threads	01018	0.40	"	"	Thread Ring Gage w/Step

8

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.3.2.1 CASE, WARHEAD

DRAWING NUMBER 65D11036
COVERING DETAIL OF DWG. 67D9700

CATEGORY	CHARACTERISTICS/DEFECTS AND TESTS	DEFECT CODE NO.	AQL OR 100%	PARAGRAPH REFERENCE REQMT.	PARAGRAPH REFERENCE TEST	METHOD OF INSPECTION
MAJOR:	Diameter aft portion of aft chamber, min.	01007	0.40	Para. 3.2	N/A	Special Plug Gage
	Concentricity of aft portion of aft chamber to forward chamber	01008	0.40	"	"	Concentricity Plug Gage
	Diameter of aft chamber, max.	01009	0.40	"	"	Special Plug Gage
	Length from rear of case to aft end of bourrelet	01010	0.40	"	"	Depth Indicator
	Diameter of bourrelet	01011	0.40	"	"	Ring Gage, Go/Not Go
	Concentricity of bourrelet to outside diameter of case	01012	0.40	"	"	Concentricity Gage w/Dial Indicator
	Outside diameter of aft section of case	01013	0.40	"	"	Adjustable Snap Gage

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.3.2.2 SLEEVE

SHEET 1 OF 1

DRAWING NUMBER 65B11040

COVERING DETAIL OF DWG. 67D9700

CATEGORY	CHARACTERISTICS/DEFECTS AND TESTS	DEFECT CODE NO.	AQL OR 100%	PARAGRAPH REFERENCE REQMT.	PARAGRAPH REFERENCE TEST	METHOD OF INSPECTION
CRITICAL:	None defined.					
MAJOR:			GROUP AQL 0.65%			
	Length	02001	0.40	Para. 3.2	N/A	Adjustable Snap Gage
	Distance across outside surface	02002	0.40	"	"	Tape Measure
	Radius of sleeve	02003	0.40	"	"	Template
	Metal defective	02004	0.40	3.3	"	Visual
MINOR:			GROUP AQL 1.00%			
	Thickness	02005	0.65	3.2	"	Indicator Thickness Gage
	Evidence of poor workmanship	02006	0.65	3.4	"	Visual

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.3.2.1 CASE, WARHEAD				SHEET 4 OF 4		DRAWING NUMBER 65D11036 COVERING DETAIL OF DWG. 67D9700
CATEGORY	CHARACTERISTICS/DEFECTS AND TESTS	DEFECT CODE NO.	AQL OR 100%	PARAGRAPH REFERENCE REQMT.	PARAGRAPH REFERENCE TEST	METHOD OF INSPECTION
MAJOR:	Length from aft end to aft portion of threads	01019	0.40	Para. 3.2	N/A	Template Gage (Length)
	Burrs or sharp edges on threads	01020	0.40	"	"	Visual
	Metal defective (See 3.3)	01021	0.40	3.3	"	Visual
MINOR:		GROUP AQL 1.00%				
	Overall length	01022	0.65	3.2	"	Length Gage
	Evidence of poor workmanship	01023	0.65	3.4	"	Visual

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.3.2.3 BASE, NOSE CONE

DRAWING NUMBER 65C11044

COVERING DETAIL OF DWG. 67D9700

CATEGORY	CHARACTERISTICS/DEFECTS AND TESTS	DEFECT CODE NO.	AQL OR 100%	PARAGRAPH REFERENCE REQMT.	PARAGRAPH REFERENCE TEST	METHOD OF INSPECTION
MINOR:		GROUP AQL 2.5%				
	Overall length	03008	0.65	Para. 3.1	N/A	Adjustable Snap Gage
	Distance from aft face to "O" ring groove	03009	0.65	"	"	Template (Length) Gage
	Depth of cavity	03010	0.65	"	"	Template (Length) Gage
	Aft face chamfer missing	03011	0.65	"	"	Visual
	Evidence of poor workmanship	03012	0.65	3.4	"	Visual

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.3.2.3 BASE, NOSE CONE — SHEET 1 OF 2 — DRAWING NUMBER 65C11044 — COVERING DETAIL OF DWG. 67D9700

CATEGORY	CHARACTERISTICS/DEFECTS AND TESTS	DEFECT CODE NO.	AQL OR 100%	PARAGRAPH REFERENCE REQMT.	PARAGRAPH REFERENCE TEST	METHOD OF INSPECTION
CRITICAL: None defined.						
MAJOR:		GROUP AQL 1.00%				
	Outside diameter	03001	0.40	Para. 3.1	N/A	Ring Gage - GO/NOT GO
	Diameter of "O" ring groove	03002	0.40	"	"	Adjustable Snap/Blade Type
	Concentricity of "O" ring groove to outside diameter	03003	0.40	"	"	Fixture w/Indicator
	Inside diameter	03004	0.40	"	"	Plug Gage-GO/NOT GO
	Concentricity of inside diameter to outside diameter	03005	0.40	"	"	Fixture w/Indicator
	Width of "O" ring groove	03006	0.40	"	"	Flat Plug Gage
	Metal defective	03007	0.40	3.3	"	Visual

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.3.2.4 NOSE CONE

DRAWING NUMBER 65C11045

COVERING DETAIL OF DWG. 67D9700

CATEGORY	CHARACTERISTICS/DEFECTS AND TESTS	DEFECT CODE NO.	AQL OR 100%	PARAGRAPH REFERENCE REQMT.	PARAGRAPH REFERENCE TEST	METHOD OF INSPECTION
MINOR:	Overall length	04007	0.65	Para. 3.1	N/A	Length Gage
	Wall thickness	04008	0.65	"	"	Dial Calipers
	Chamfer missing on aft end	04009	0.65	"	"	Length Gage
	Evidence of poor workmanship	04010	0.65	3.4	"	Visual

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH		SHEET 1 OF 2				DRAWING NUMBER
4.3.2.4	NOSE CONE					65C11045
						COVERING DETAIL OF DWG.
						67D9700
CATEGORY	CHARACTERISTICS/DEFECTS AND TESTS	DEFECT CODE NO.	AQL OR 100%	PARAGRAPH REFERENCE REQMT.	PARAGRAPH REFERENCE TEST	METHOD OF INSPECTION
CRITICAL:	None defined.					
MAJOR:		GROUP AQL 0.65%				
	Outside diameter of aft section	04001	0.40	Para. 3.1	N/A	Ring Gage, GO/NOT GO
	Length of aft section	04002	0.40	"	"	Template (Length) Gage
	Perpendicularity of aft section to diameter of aft section (wall to base area)	04003	0.40	"	"	Ring Gage w/Feeler
	Straightness of outside surface	04004	0.40	"	"	Fixture w/Indicator
MINOR:		GROUP AQL 4.0%				
	Outside diameter	04005	0.65	"	"	Ring Gage, GO/NOT GO
	Concentricity of outside diameter to diameter of aft section	04006	0.65	"	"	Concentricity Gage w/Dial Indicator

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH		SHEET 1 OF 1				DRAWING NUMBER 65B11042
4.3.2.6	PIN, CONE					COVERING DETAIL OF DWG. 67D9700
CATEGORY	CHARACTERISTICS/DEFECTS AND TESTS	DEFECT CODE NO.	AQL OR 100%	PARAGRAPH REFERENCE REQMT.	PARAGRAPH REFERENCE TEST	METHOD OF INSPECTION
CRITICAL: None defined.						
MAJOR:		GROUP AQL 0.65%				
	Length, overall	06001	0.40	Para. 3.1	N/A	Adjustable Snap Gage
	Diameter, body	06002	0.40	"	"	Ring Gage, GO/NOT GO
	Diameter, ends	06003	0.40	"	"	Ring Gage, GO/NOT GO
MINOR:		GROUP AQL 0.65%				
	Evidence of poor workmanship	06004	0.65	3.4	"	Visual

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.3.2.5 CONE ASSEMBLY, NOSE

SHEET 1 OF 1

DRAWING NUMBER 65C11043

COVERING DETAIL OF DWG. 67D9700

CATEGORY	CHARACTERISTICS/DEFECTS AND TESTS	DEFECT CODE NO.	AQL OR 100%	PARAGRAPH REFERENCE REQMT.	PARAGRAPH REFERENCE TEST	METHOD OF INSPECTION
CRITICAL:	None defined.					
MAJOR:			GROUP AQL 0.40%			
	Parts not firmly affixed	05001	0.40	Para. 3.1	N/A	Manual
	Adhesive on outside surface	05002	0.40	"	"	Visual
MINOR:			GROUP AQL 0.65%			
	Evidence of poor workmanship	05003	0.65	3.4	"	Visual

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH				SHEET 1 OF 1		DRAWING NUMBER 65B11039
4.3.2.8	PLATE, PUSHER					COVERING DETAIL OF DWG. 67D9700
CATEGORY	CHARACTERISTICS/DEFECTS AND TESTS	DEFECT CODE NO.	AQL OR 100%	PARAGRAPH REFERENCE REQMT.	TEST	METHOD OF INSPECTION
CRITICAL: None defined.						
MAJOR:			GROUP AQL 0.40%			
	Outside diameter	08001	0.40	Para. 3.1	N/A	Snap Gage
MINOR:			GROUP AQL 1.00%			
	Thickness	08002	0.65	"	"	Snap Gage
	Evidence of poor workmanship	08003	0.65	3.4	"	Visual

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.3.2.7 SPACER | SHEET 1 OF 1 | DRAWING NUMBER 65B11038 | COVERING DETAIL OF DWG. 67D9700

CATEGORY	CHARACTERISTICS/DEFECTS AND TESTS	DEFECT CODE NO.	AQL OR 100%	PARAGRAPH REFERENCE REQMT.	PARAGRAPH REFERENCE TEST	METHOD OF INSPECTION
CRITICAL:	None defined.					
MAJOR:	None defined.					
MINOR:			GROUP AQL 1.5%			
	Outside diameter	07001	0.65	Para. 3.1	N/A	Snap Gage
	Thickness	07002	0.65	"	"	Snap Gage
	Evidence of poor workmanship	07003	0.65	3.4	"	Visual

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	FLECHETTE, 20 GRAIN, MIL-STD-105D, LEVEL S-1, TABLE IIA SHALL BE USED FOR THE FOLLOWING	SHEET 1 OF 1	DRAWING NUMBER 67D9701
4.3.2.9	CHARACTERISTICS/DEFECTS (EXCEPT AS NOTED)		COVERING DETAIL OF DWG 67D9700

CATEGORY	CHARACTERISTICS/DEFECTS AND TESTS	DEFECT CODE NO.	AQL OR 100%	PARAGRAPH REFERENCE REQMT.	PARAGRAPH REFERENCE TEST	METHOD OF INSPECTION
CRITICAL: None defined.						
MAJOR:	GROUP AQL 0.65% (Codes 09001, 09002, 09003 Only)					
	Length	09001	0.40	Para. 3.1	N/A	Adjustable Snap Gage
	Length from center of flechette to outside of fin	09002	0.40	"	"	Location Gage
	Overall fin width	09003	0.40	"	"	Adjustable Snap Gage
	Hardness Test	09004	----	67D9701	4.4.1.1	Hardness Tester
	Fin Bend Test	09005	----	67D9701	4.4.2	Fixture

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	FLECHETTE, 20 GRAIN MIL-STD-105D, LEVEL S-1, TABLE IIA SHALL BE USED FOR THE FOLLOWING CHARACTERISTICS/DEFECTS (EXCEPT AS NOTED)			SHEET 2 OF 2		DRAWING NUMBER 67D9701
4.3.2.9						COVERING DETAIL OF DWG. 67D9700
CATEGORY	CHARACTERISTICS/DEFECTS AND TESTS	DEFECT CODE NO.	AQL OR 100%	PARAGRAPH REFERENCE REQMT.	TEST	METHOD OF INSPECTION
MINOR:		GROUP AQL 1.5%				
	Angle between fins	09006	0.65	Para. 3.1	N/A	Comparator & Layout
	Fin surfaces not parallel to center axis	09007	0.65	"	"	Fixture with Dial Indicator
	Flash on point	09008	0.65	"	"	Visual
	Evidence of poor workmanship	09009	0.65	3.4	"	Visual

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	FLECHETTE, 60 GRAIN, MIL-STD-105D, LEVEL S-1, TABLE IIA SHALL BE USED F/THE FOLLOWING	SHEET 1 OF 2	DRAWING NUMBER 9260010
4.3.2.10	CHARACTERISTICS/DEFECTS (EXCEPT AS NOTED)		COVERING DETAIL OF DWG. 9259995

CATEGORY	CHARACTERISTICS/DEFECTS AND TESTS	DEFECT CODE NO.	AQL OR 100%	PARAGRAPH REFERENCE REQMT.	PARAGRAPH REFERENCE TEST	METHOD OF INSPECTION
CRITICAL:	None defined.					
MAJOR:	GROUP AQL 0.65% (Codes 10001, 10004, 10005 Only)					
	Length	10001	0.40	Para. 3.1	N/A	Adjustable Snap Gage
	Hardness Test	10002	----	9260010	4.4.1.2	Hardness Tester
	Fin Bend Test	10003	----	9260010	4.4.2	Fixture
	Length from center of flechette to outside of fin	10004	6.50	3.1	N/A	Location Gage
	Overall Fin Width	10005	6.50	"	"	Adjustable Snap Gage

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	FLECHETTE, 60 GRAIN, MIL-STD-105D, LEVEL S-1, TABLE IIA SHALL BE USED F/THE FOLLOWING	SHEET 2 OF 2	DRAWING NUMBER 9260010
4.3.2.10	CHARACTERISTICS/DEFECTS (EXCEPT AS NOTED)		COVERING DETAIL OF DWG. 9259995

CATEGORY	CHARACTERISTICS/DEFECTS AND TESTS	DEFECT CODE NO.	AQL OR 100%	PARAGRAPH REFERENCE REQMT.	PARAGRAPH REFERENCE TEST	METHOD OF INSPECTION
MINOR:		GROUP AQL 1.5%				
	Angle between fins	10006	0.65	Para. 3.1	N/A	Comparator & Chart
	Fin surfaces not parallel to center axis	10007	0.65	"	"	Fixture with Dial Indicator
	Flash on point	10008	0.65	"	"	Visual
	Evidence of poor workmanship	10009	0.65	3.4	"	Visual

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	FLECHETTE BELT ASSEMBLY 20 GRAIN			SHEET 1 OF 1		DRAWING NUMBER 9257226
4.3.2.11						COVERING DETAIL OF DWG. 67D9700
CATEGORY	CHARACTERISTICS/DEFECTS AND TESTS	DEFECT CODE NO.	AQL OR 100%	PARAGRAPH REFERENCE REQMT.	TEST	METHOD OF INSPECTION
CRITICAL:	None defined.					
MAJOR:	None defined.					
MINOR:			GROUP AQL 0.65%			
	Evidence of poor workmanship	11001	0.65	Para. 3.4	N/A	Visual

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	FLECHETTE BELT ASSEMBLY (60 GRAIN)			SHEET 1 OF 1		DRAWING NUMBER 9260011 COVERING DETAIL OF DWG. 9259995
4.3.2.12						
CATEGORY	CHARACTERISTICS/DEFECTS AND TESTS	DEFECT CODE NO.	AQL OR 100%	PARAGRAPH REFERENCE REQMT.	PARAGRAPH REFERENCE TEST	METHOD OF INSPECTION
CRITICAL:	None defined.					
MAJOR:	None defined.					
MINOR:	Evidence of poor workmanship	12001	0.65	Para. 3.4	N/A	Visual

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4.4 Testing

4.4.1 Hardness Test of Flechette (See applicable subparagraph).

4.4.1.1 Hardness Test of Flechette, 20 Grain (See Dwg. 67C9701) - Major Defect.-During the heat treatment process of the Flechettes, two (2) samples shall be selected each hour for hardness testing. The test shall be performed as specified in 4.6.1, using equipment in accordance with 4.5. Any Flechette sample that does not meet the minimum hardness requirement shall be considered defective. If either sample fails to pass hardness requirements, the Flechettes which are represented by this sample shall be rejected.

4.4.1.2 Hardness Test of Flechette, 60 Grain (See Dwg. 9260010) - Major Defect.-During the heat treatment process of the Flechettes, two (2) samples shall be selected each hour for hardness testing. The test shall be performed as specified in 4.6.1, using equipment in accordance with 4.5. Any Flechette sample that does not meet the minimum hardness requirement shall be considered defective. If either sample fails to pass hardness requirements, the Flechettes which are represented by this sample shall be rejected. The Flechettes used for this test are to be used for Fin Bend Test (See 4.4.2).

4.4.2 Fin Bend Test (See Dwg. No. 9260010) - Major Defect.- The Flechettes selected each hour and used for test in paragraph 4.4.1.2 shall be subjected to this test. Any Flechette which does not meet requirements shall be considered defective. If either unit of the sample fails, the Flechettes which are represented by this sample shall be rejected. This is considered a Destructive Test.

4.5 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.3.2 and 4.6 (See 6.3).

4.6 Test Methods and Procedures

4.6.1 Hardness Test (See 4.4.1.1 and 4.4.1.2).-A specimen shall be prepared from each Flechette in the sample. Each specimen shall be prepared and tested in accordance with the Federal Test Method Standard No. 151, Method 243.1.

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4.6.1.1 Retest.-The lot may be reheated and retested.

4.6.2 Fin Bend Test (See 4.4.2).-Each Flechette shall be held firmly in a fixture and one fin shall be bent a minimum of fifteen degrees (15°) radially.

5. PREPARATION FOR DELIVERY

5.1 Packaging and Marking - Level C.-The warhead shall be packaged and marked in accordance with Military Specification MIL-STD-1169.

5.2 Packing and Marking - Level C.-The packaged warheads shall be packed and marked in accordance with Military Specification MIL-STD-1169.

6. NOTES

6.1 Intended Use.-See paragraph 1.1 of this specification.

6.2 Ordering Data.-Procurement documents shall specify the following:

a. Title, number and date of this and the item detail specifications.

b. Data Cards.-Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167, except when propellant or explosive lots are involved. Acceptance and Description Sheets shall be prepared in accordance with MIL-STD-1171.

c. Quantity required and delivery schedules.

d. Serialization requirements, if applicable.

e. Production lot acceptance, if other than specified in Section 4 of this specification.

f. First Article Sample requirements, if other than specified in Section 4 of this specification.

g. Packaging requirements, if other than specified in Section 5 of this specification.

h. Inspection equipment design requirements (See 4.5 and 6.3).

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6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) Designs and Contractor Designs. SIE Designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the Contractor. The Contractor need not furnish any design when a complete Government SIE Design is provided. Unless otherwise specified, however, the Contractor may submit alternate or modified Contractor Designs of SIE in accordance with paragraphs 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE Designs.-SIE Designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor Designs.-Contractor Designs are required for all inspection equipment for which SIE Designs are not specified, and may include commercial equipment which the Contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor Designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government, and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE Designs, if not otherwise covered thereby (See paragraph 6.3.1c.). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The Contractor is referred to MIL-HDBK-204, "Inspection Equipment Design", for guidance. The specification number and other specific identifying information from Section 4 of this specification shall be referenced on each Contractor Design, together with the component or assembly drawing number and revision letter to which the specific design applies.

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6.3.3 Submission of Designs for Approval.-Contractor Designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SMUPA-QA-A-R, Dover, New Jersey, 07801, in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be normally accomplished within one month after receipt.

6.4 Submission of First Article Sample.-Instructions as to the submission and location for testing and evaluation of the first article sample shall be obtained from the Contracting Officer. Upon receipt of such request, the Contracting Officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to:

Commander
Picatinny Arsenal
ATTN: SMUPA-QA-I-Q
Dover, New Jersey 07801

6.5 Definitions

6.5.1 Contractor.-Unless otherwise specified, the word "Contractor", used throughout this specification, shall mean any supplier or producer of items or material to the Government. This shall include commercial contractors, sub-contractors, Government-Owned, Contractor-Operated (GOCO) and Government-Owned, Government-Operated (GOGO) plants.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1340-A386

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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